Spin-Polarization Control of Photoelectrons Using Poincaré Fields

(Extra notes)

Constants

- The normalized field amplitude tells the strength of the field relative to the electron rest energy

$$a_0 = \frac{eE}{m_e c\omega} \tag{1}$$

- Keldysh parameter tells the regime, tunnel or multiphoton

$$\gamma = \frac{\omega}{E}\sqrt{2m_e I_p} = \sqrt{\frac{2I_p}{m_e a_0^2 c}} \tag{2}$$

- The ionization potential

$$I_p = m_e c^2 - E, \qquad \text{with} \qquad E = m_e c^2\sqrt{1-(Z\alpha^2)} \tag{3}$$

if $Z\alpha \ll 1$, (nonrelativistic case),

$$I_p \sim \frac{m_e Z\alpha^2 c}{2} \tag{4}$$

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(Extra notes)

Vector spherical harmonics and the magnetic multipole field

- The multipole order electromagnetic field is written, the poles being magnetic

$$\boldsymbol{A}_{jm}(\boldsymbol{k},t) = \mathrm{Re}\, A_0 \boldsymbol{\Phi}_{jm}\left(\theta_{\boldsymbol{k}},\varphi_{\boldsymbol{k}}\right)\delta(|\boldsymbol{k}|-\omega)e^{i\omega t}, \tag{5}$$

$$\boldsymbol{A}_{jm}(\boldsymbol{r},t) = \mathrm{Re}\, A_0 g_j(kr)\boldsymbol{\Phi}_{jm}\left(\theta,\varphi\right)e^{i\omega t}, \tag{6}$$

where $\boldsymbol{\Phi}_{jm}\left(\theta,\varphi\right) = \hat{\boldsymbol{r}} \times \boldsymbol{\nabla} Y_{jm}\left(\theta,\varphi\right)$ and $g_j(z) = \sqrt{\frac{\pi}{2z}}J_{j+1/2}(z)$ is the Bessel function of the first kind

$$j \geqslant 1, \qquad\qquad |m| < j-1 \tag{7}$$

j	m	Structure	Interpretation
1	0	(dipole) Simplest radiative field	Common in laser, no azymuthal twsit, linear polarized dipole.
2	± 1	(quadrupole) Four-lobbed pattern	Forbidden transitions, nuclear EM fields, Circular polariz., spin angular moment.
3	± 2	(octupole) Eight-lobbed pattern	High energy transit., structured beams, nontrivial orbital structure
$4+$	± 3	Higher multipoles	HHG, nuclear EM moments, astrophysics, high order vortex beams etc..

- j is used not as the sum of ℓ and spin s, but rather as a convention.
- Notice that the inequality (7) is strict

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(Extra notes)

Calculation of helicity

- Using the metric $(+,-,-,-)$ and having $\boldsymbol{\sigma}$ as the Pauli matrices

$$\gamma^0 \equiv \beta = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix}, \qquad \gamma = \begin{pmatrix} 0 & \boldsymbol{\sigma} \\ -\boldsymbol{\sigma} & 0 \end{pmatrix}, \qquad \alpha = \gamma^0 \boldsymbol{\gamma}, \qquad \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\sigma} & 0 \\ 0 & \boldsymbol{\sigma} \end{pmatrix}. \tag{8}$$

- The helicity of the particle is the projection of its spin along $\boldsymbol{n} = \boldsymbol{p}/|\boldsymbol{p}|$ and is a "good quantum number"
- Helicity density is expressed as

$$\zeta(\boldsymbol{p}) = \langle \widetilde{\psi}(\boldsymbol{p})| \left(\boldsymbol{n} \cdot \frac{1}{2} \boldsymbol{\Sigma} \right) |\widetilde{\psi}(\boldsymbol{p})\rangle, \qquad \boldsymbol{n} = (\sin\theta \sin\varphi, \sin\theta\cos\varphi, \cos\varphi), \tag{9}$$

$$\zeta(\boldsymbol{p}) = \frac{1}{2} \int d^3\boldsymbol{p}\, \widetilde{\psi}^*(\boldsymbol{p}) \begin{pmatrix} \cos\theta & \sin\theta e^{-i\varphi} & 0 & 0 \\ \sin\theta e^{i\varphi} & -\cos\theta & 0 & 0 \\ 0 & 0 & \cos\theta & \sin\theta e^{-i\varphi} \\ 0 & 0 & \sin\theta e^{i\varphi} & -\cos\theta \end{pmatrix} \widetilde{\psi}(\boldsymbol{p}) \tag{10}$$

- The case of a free electron in its $\gamma = +1/2$ helicity state is in the supp. materials

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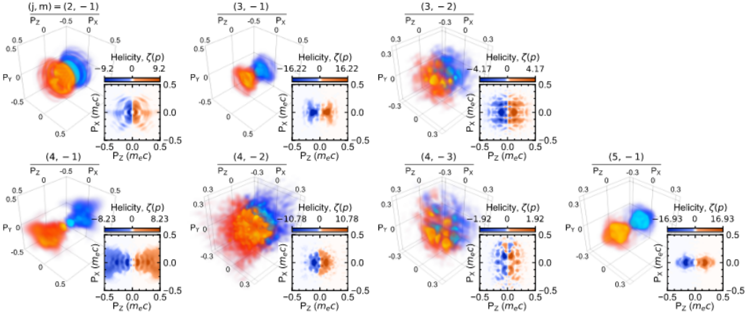
(Extra notes)

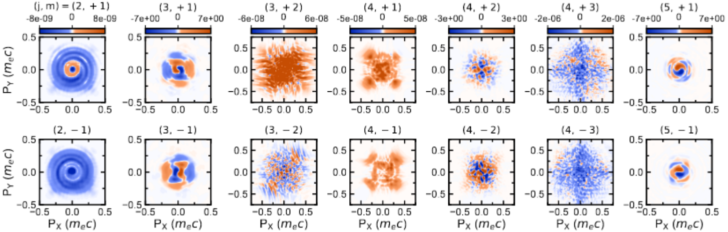
Helicity density $\zeta(p)$ of Ne^{9+} - Single mode ionization case - $(m < 0)$

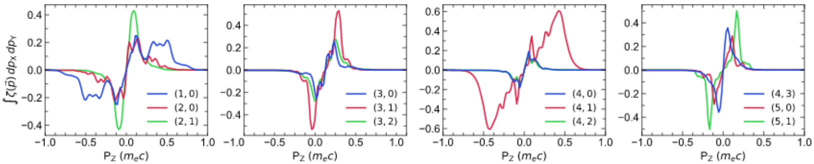


Spin-Polarization Control of Photoelectrons Using Poincaré Fields

(Extra notes)

Helicity density $\zeta(p)$ of Ne^{9+} - Single mode ionization case - Integrated





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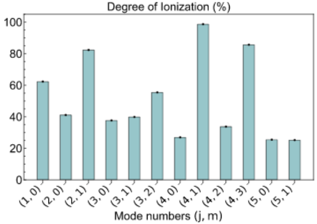
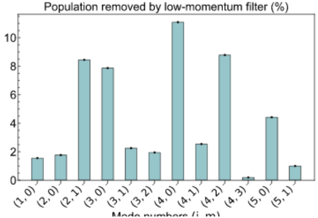
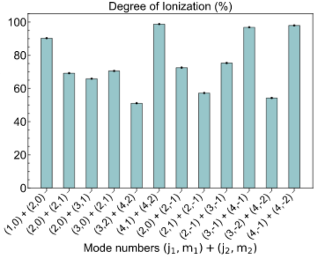
(Extra notes)

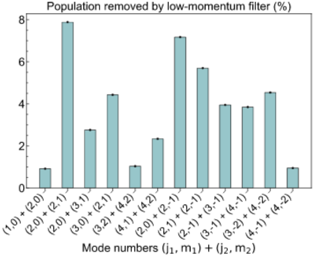
Visual effect of the mask function $\mathcal{M}_\mu(\boldsymbol{r})$





- Left : Degree of ionization for each mode, bound population was removed by using $\mathcal{M}_\mu(\boldsymbol{r})$
- Right : Percent of the population removed by the low momentum filter $\mathcal{M}_\mu(\boldsymbol{p})$





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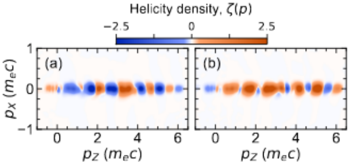
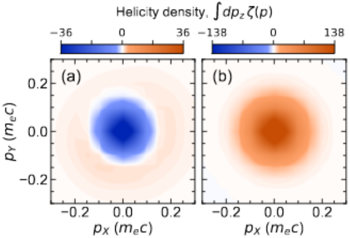
(Extra notes)

Helicity density $\zeta(p)$ of Ar^{17+}





- Irradiation by a left-(a) and right-(b) -circularly-polarized field
- Leftmost panels : Distribution in the xy-plane
- Rightmost panels : Distribution integrated along the z momentum direction